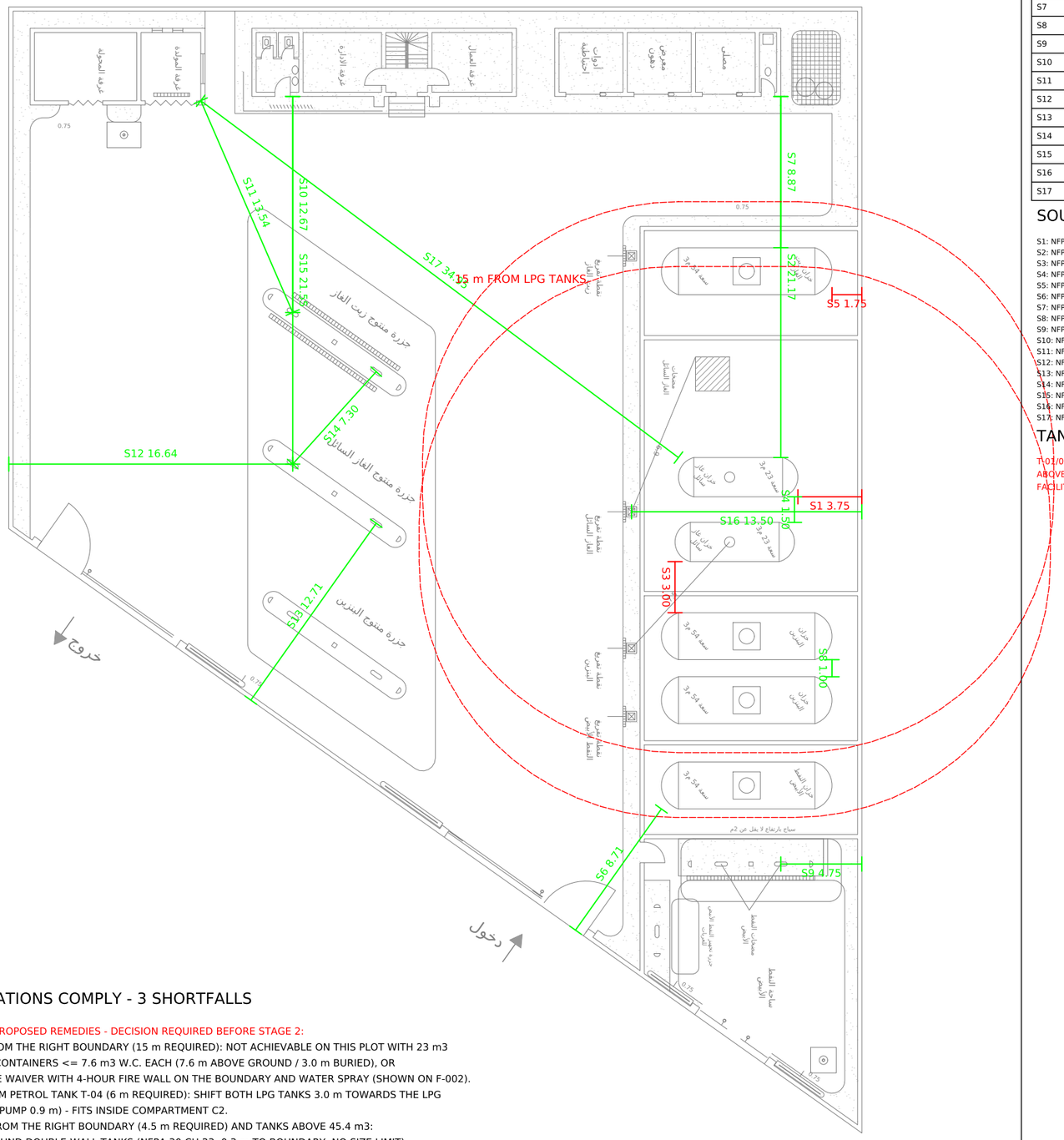
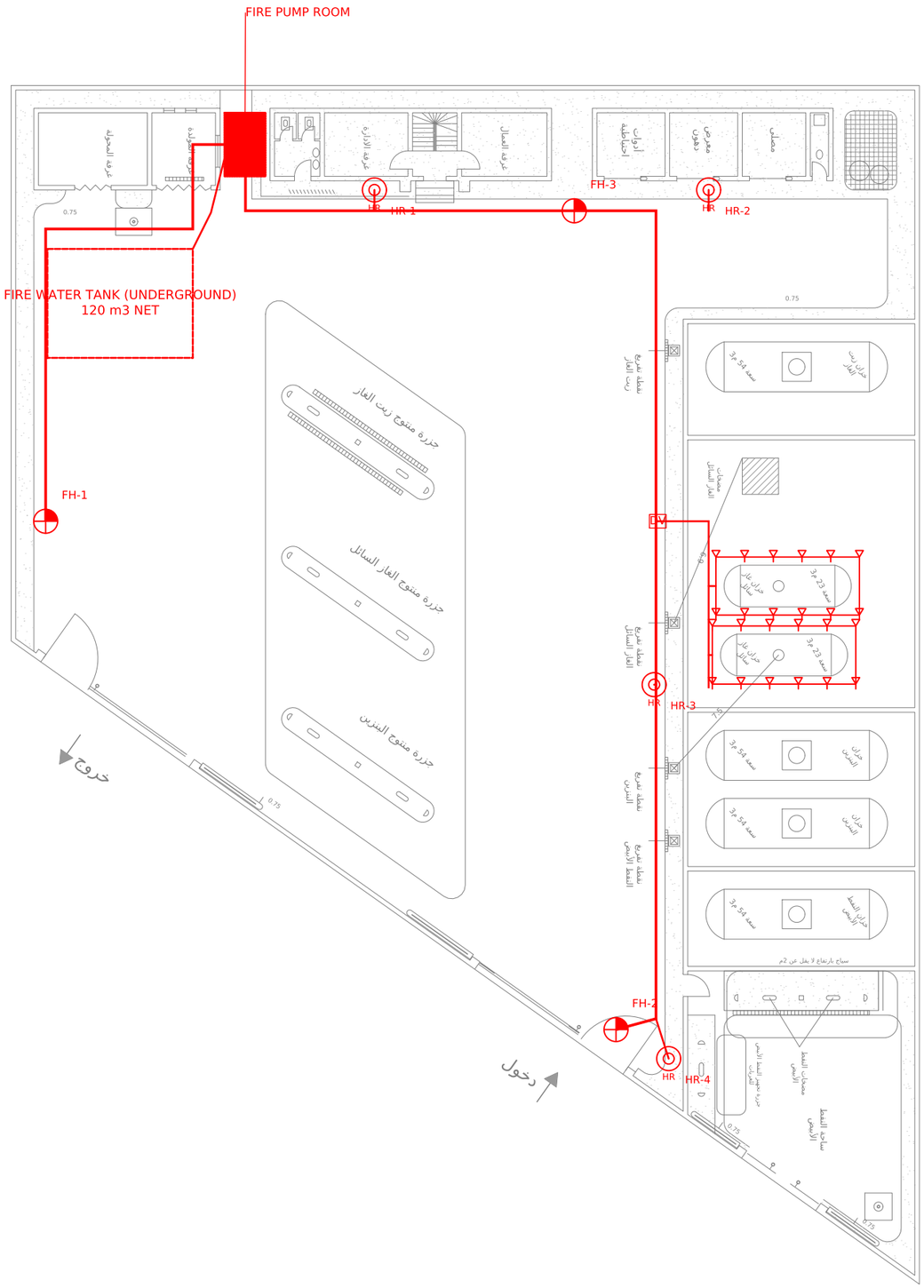

15 m FROM LPG TANKS
S1 3.75
S2 21.17
S3 3.00
S4 1.50
S5 1.75
S6 8.71
S7 8.87
S8 11.00
S9 4.75
S10 12.67
S11 13.54
S12 16.64
S13 12.71
S14 7.30
S15 21.55
S16 13.50
S17 34.33
0.75
0.75
0.75



FIRE WATER DEMAND

LPG TANKS SURFACE (2 TANKS)	95.1 m2
WATER SPRAY 10.2 L/min/m2	970 L/min
HYDRANT HOSE STREAMS (2 x 475)	950 L/min
DESIGN FLOW	1920 L/min
DURATION	60 min
REQUIRED VOLUME	115.2 m3
FIRE TANK (NET)	120 m3

FIRE PUMPS (NFPA 20):
- MAIN ELECTRIC 2000 L/min @ 70 m
- STANDBY DIESEL 2000 L/min @ 70 m
- JOCKEY 50 L/min @ 80 m

SPECIFICATION

1. FIRE MAIN: HDPE PE100 PN16, 110 mm, BURIED 0.9 m MIN.; DUCTILE IRON ABOVE GROUND.
2. HYDRANTS: PILLAR TYPE 2 x 65 mm INSTANTANEOUS OUTLETS (IRAQI CIVIL DEFENCE COUPLING), ISOLATING VALVE ON EACH BRANCH.
3. HOSE REELS: 25 mm x 30 m, IN CABINETS.
4. LPG TANK COOLING: AUTOMATIC DELUGE VALVE (DV) OUTSIDE COMPARTMENT C2, OPEN NOZZLES ON A RING 0.45 m CLEAR OF EACH TANK, 10.2 L/min/m2 OVER THE WHOLE TANK SURFACE (NFPA 15).
5. DV OPENED BY HEAT/FLAME DETECTION OVER THE LPG TANKS AND BY MANUAL RELEASE AT THE DV.
6. FIRE TANK: RC, WATERPROOFED, TRAFFIC-RATED ROOF, FILLED FROM THE MAINS (P-001), WITH A CIVIL DEFENCE INLET BREECHING AT THE FRONT.
7. SEE F-006 FOR THE PUMP SET AND RISER DIAGRAM.

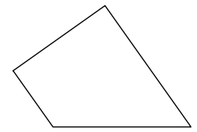
CIVIL DEFENCE

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GENERAL NOTES

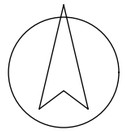
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REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

FIRE WATER NETWORK

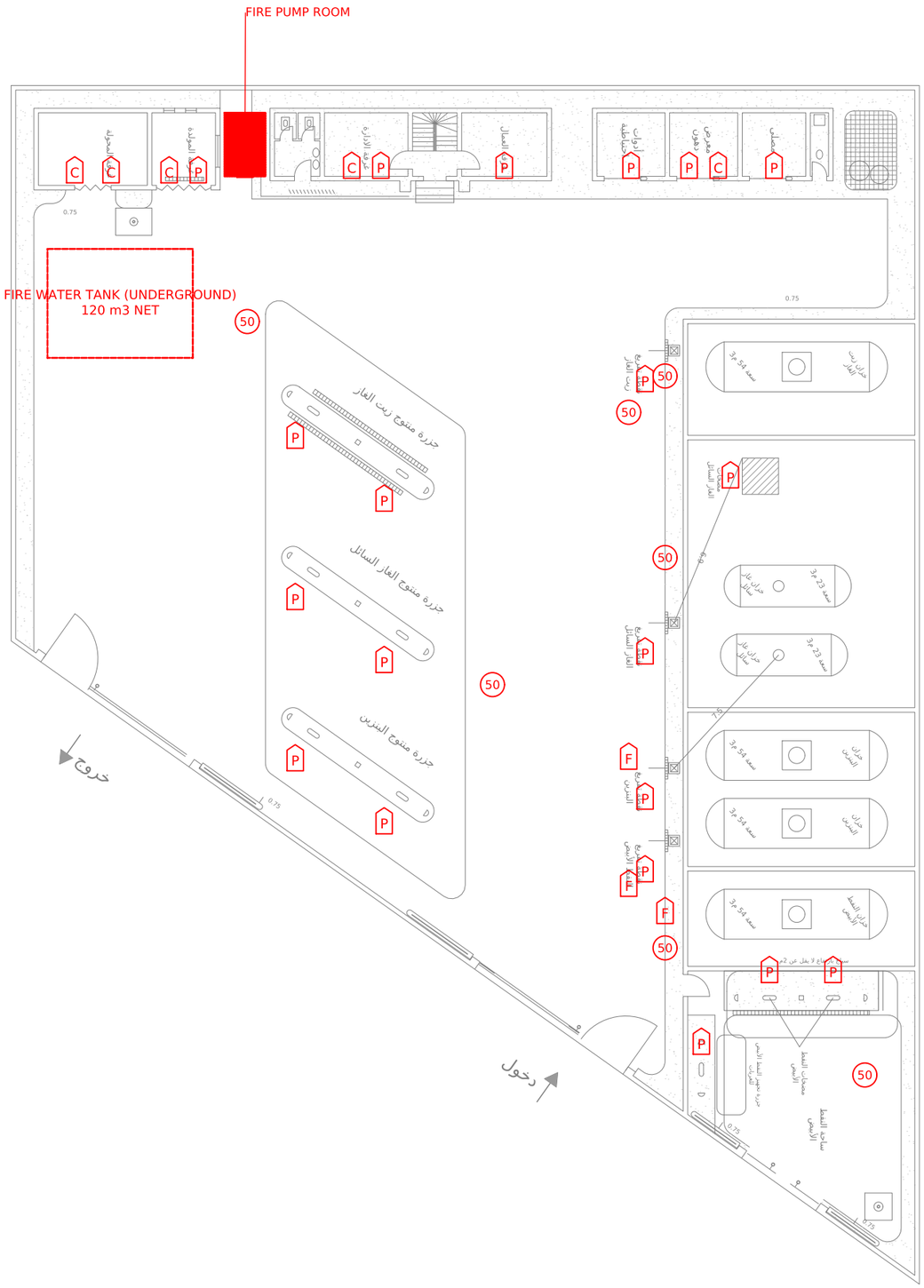
HYDRANTS, HOSE REELS, LPG SPRAY COOLING

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE	DATE
1:300	18/10/2026
PHASE: DESIGN DEVELOPMENT	STATUS: FOR REVIEW

DRAWING No.
F-002

SHEET
02
OF 8



LOCATION	TYPE	QTY
ISLAND I-1 (GAS OIL)	FE-P	2
ISLAND I-2 (LPG)	FE-P	2
ISLAND I-3 (GASOLINE)	FE-P	2
ISLAND I-4 (KEROSENE)	FE-P	2
ISLAND I-5 (KEROSENE (CARTS))	FE-P	1
ISLANDS AREA (CANOPY)	FE-W	2
KEROSENE YARD	FE-W	1
UNLOADING POINT - GAS OIL	FE-P	1
UNLOADING POINT - LPG	FE-P	1
UNLOADING POINT - GASOLINE	FE-P	1
UNLOADING POINT - KEROSENE	FE-P	1
UNLOADING POINTS	FE-W	1
UNLOADING POINTS	FE-F	2
TANK COMPARTMENT C1	FE-W	1
TANK COMPARTMENT C2 (LPG)	FE-W	1
TANK COMPARTMENT C3	FE-W	1
TANK COMPARTMENT C3	FE-F	1
LPG PUMP	FE-P	1
TRANSFORMER ROOM	FE-C	2
GENERATOR ROOM	FE-C	1
GENERATOR ROOM	FE-P	1
ADMINISTRATION (MDB)	FE-C	1
ADMINISTRATION	FE-P	1
WORKERS ROOM	FE-P	1
SPARE PARTS SHOP	FE-P	1
PAINTS SHOWROOM	FE-P	1
PAINTS SHOWROOM	FE-C	1
PRAYER ROOM	FE-P	1
FIRE PUMP ROOM	FE-P	1

TOTALS

FE-C PORTABLE EXTINGUISHER, CO2 6 kg: 5
FE-F PORTABLE EXTINGUISHER, AFFF FOAM 9 L: 3
FE-P PORTABLE EXTINGUISHER, DRY POWDER 12: 21
FE-W WHEELED EXTINGUISHER, DRY POWDER 50 : 7

NOTES

1. MOUNT PORTABLE UNITS WITH THE TOP <= 1.5 m ABOVE FLOOR, IN CABINETS OUTDOORS.
2. MAX. TRAVEL DISTANCE TO A UNIT 15 m (CLASS B).
3. CO2 ONLY IN ELECTRICAL ROOMS; NO WATER TYPE NEAR ELECTRICAL EQUIPMENT.
4. INSPECT MONTHLY, SERVICE YEARLY (NFPA 10).

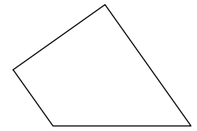
CIVIL DEFENCE

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GENERAL NOTES

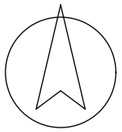
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PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

PORTABLE EXTINGUISHERS

LAYOUT AND QUANTITY SCHEDULE

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
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DATE
18/10/2026

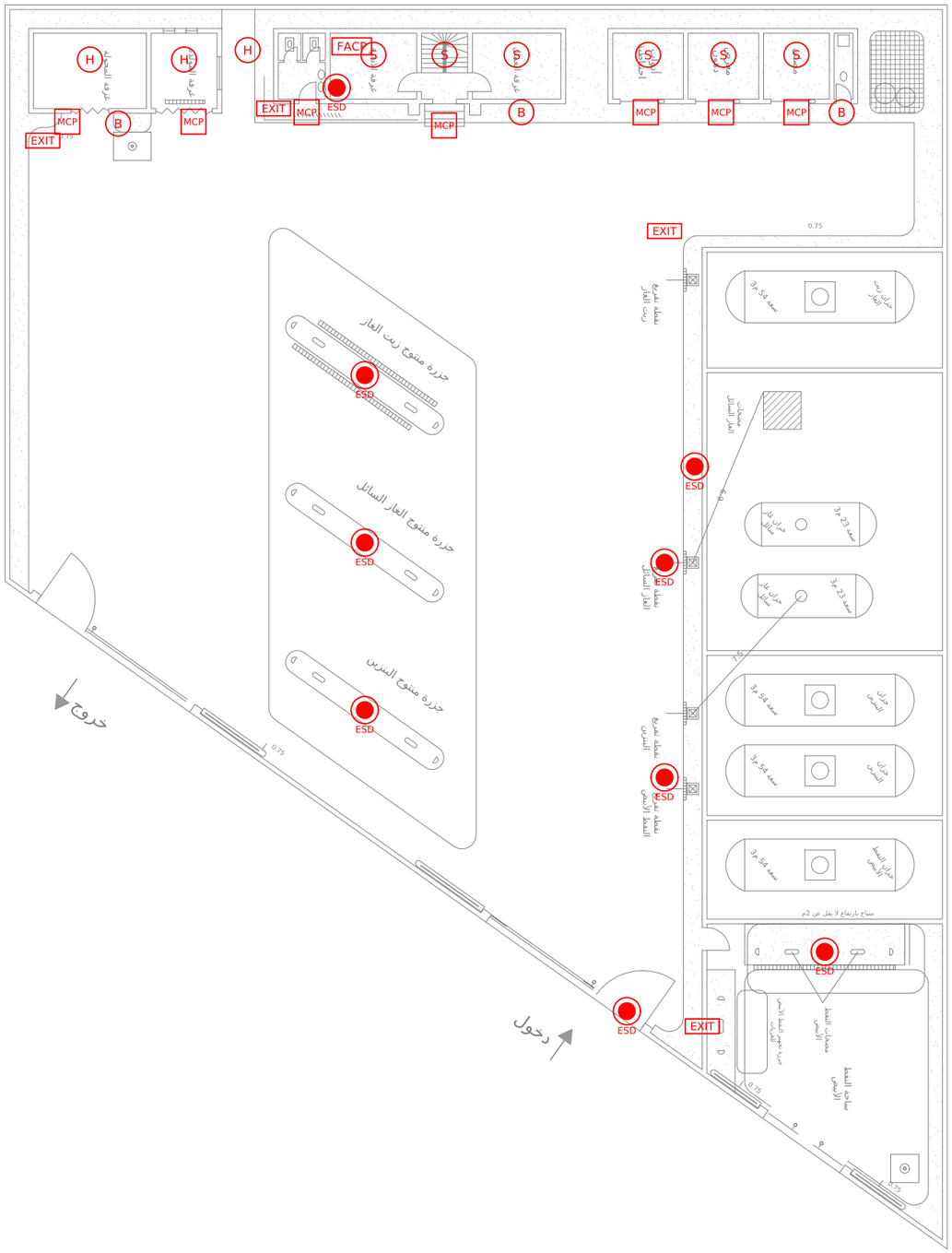
PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
F-003

SHEET
03
OF 8

FACP IN ADMINISTRATION



0 3 6 9 12 15m

SCALE 1:300 (A3)

FIRE ALARM

CONVENTIONAL 4-ZONE PANEL (NFPA 72 / EN 54):
Z1 BUILDING 1 (TRANSFORMER, GENERATOR)
Z2 BUILDING 2 (WC, ADMIN, STAIR, WORKERS)
Z3 BUILDING 3 (SHOPS, PRAYER, WC)
Z4 FIRE PUMP ROOM + LPG DELUGE STATUS
CABLE: FIRE-RESISTANT 2 x 1.5 mm2 (PH120).
STANDBY BATTERIES 24 h + 30 min ALARM.

EMERGENCY SHUT-DOWN (ESD)

9 RED MUSHROOM PUSH BUTTONS (STAY-PUT):
AT EVERY ISLAND, THE UNLOADING POINTS, THE
LPG COMPARTMENT, THE OFFICE AND THE GATE.
ANY BUTTON TRIPS: ALL DISPENSERS, SUBMERSIBLE
AND TRANSFER PUMPS, LPG PUMP AND THE CANOPY
POWER (NFPA 30A 6.7). FIRE PUMPS, FIRE ALARM
AND EMERGENCY LIGHTING STAY ON.
RESET ONLY AT THE PANEL IN THE OFFICE.
BUTTONS 6 - 30 m FROM DISPENSERS, IN VIEW.

SAFETY SIGNS (ARABIC + ENGLISH)

NO SMOKING / NO NAKED FLAME - AT EVERY ISLAND,
THE TANK AREA AND THE UNLOADING POINTS.
STOP ENGINE - SWITCH OFF MOBILE PHONES.
LPG - FLAMMABLE GAS (TANK COMPARTMENT C2).
EMERGENCY SHUT-DOWN - AT EVERY ESD BUTTON.
FIRE EXIT / ASSEMBLY POINT - SEE F-005.

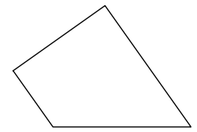
CIVIL DEFENCE

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GENERAL NOTES

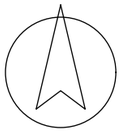
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PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

FIRE ALARM AND ESD

DETECTION, CALL POINTS, SHUT-DOWN

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
1:300

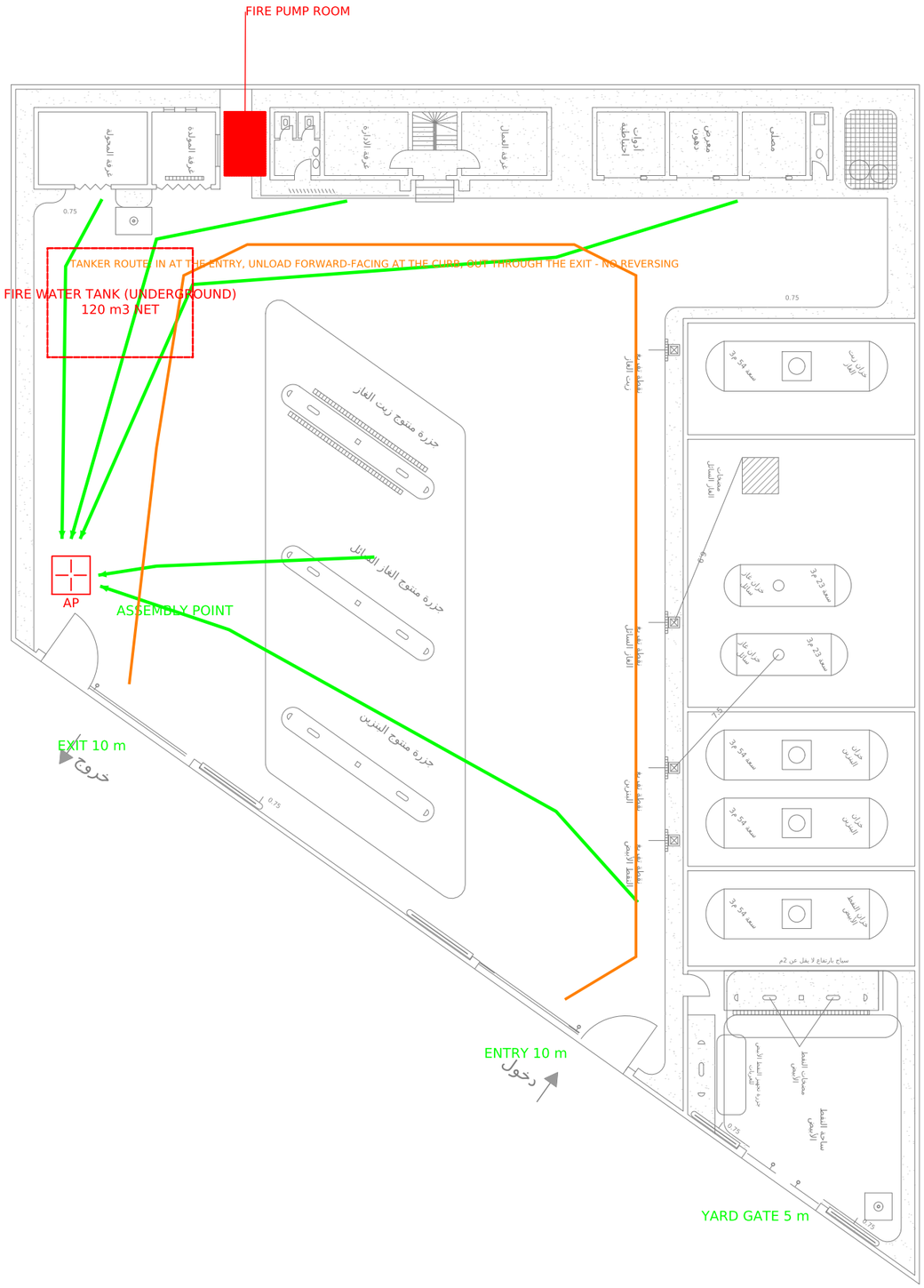
DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
F-004

SHEET
04
OF 8



SCALE 1:300 (A3)

EVACUATION

1. ON ALARM: STOP ALL DISPENSING (ESD), STOP UNLOADING, CLOSE TANKER VALVES.
2. STAFF AND CUSTOMERS WALK TO THE ASSEMBLY POINT AT THE FRONT-LEFT CORNER, AWAY FROM THE TANKS (> 30 m), AS SHOWN.
3. GATES STAY OPEN FOR CIVIL DEFENCE VEHICLES; CLEAR WIDTH 10 m AT ENTRY AND EXIT.
4. A TANKER UNLOADS ONLY WHEN THE STATION ATTENDANT, EXTINGUISHERS AND THE STATIC EARTH CLAMP ARE IN PLACE (E-009).
5. ROUTES ARE PAVED, LIT (E-004) AND KEPT FREE OF PARKED VEHICLES.

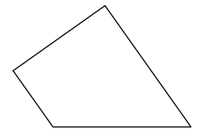
CIVIL DEFENCE

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GENERAL NOTES

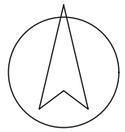
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PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

EVACUATION PLAN

EXITS, ROUTES, TANKER CIRCULATION

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
1:300

DATE
18/10/2026

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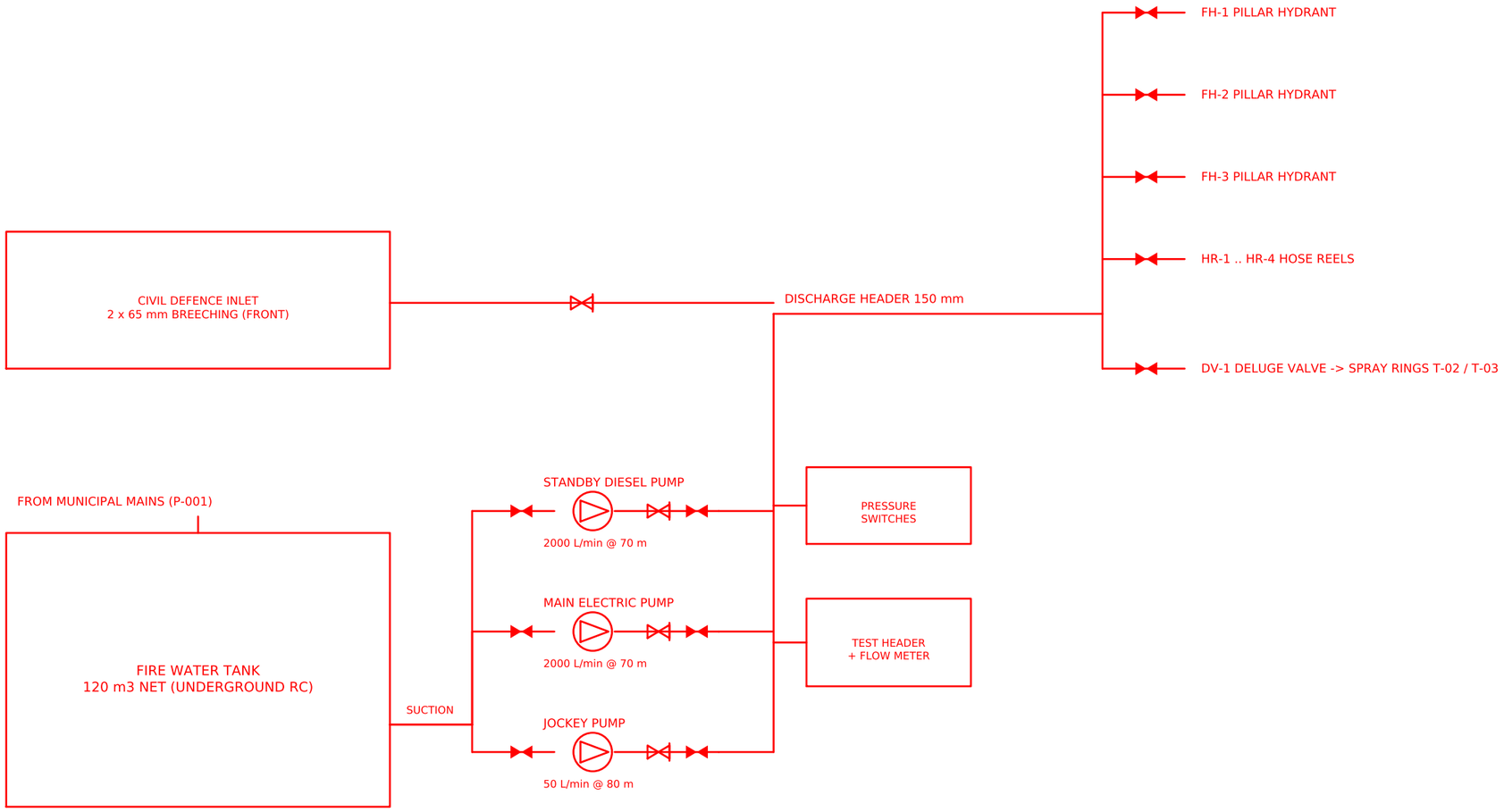
STATUS
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DRAWING No.
F-005

SHEET
05
OF 8

FIRE WATER SYSTEM - SCHEMATIC (NOT TO SCALE)

1. PUMP SET TO NFPA 20: JOCKEY STARTS AT 7.0 bar, MAIN AT 6.0 bar, DIESEL AT 5.5 bar; MAIN AND DIESEL STOP MANUALLY ONLY.
2. DIESEL PUMP WITH ITS OWN 8 h FUEL TANK AND TWO BATTERY SETS; EXHAUST LED OUTSIDE.
3. MAIN PUMP FED FROM THE MDB AHEAD OF THE MAIN BREAKER AND FROM THE GENERATOR (E-007).
4. DESIGN FLOW 1920 L/min FOR 60 min = 115.2 m3; TANK 120 m3 NET.
5. ALL VALVES LOCKED OPEN; PUMP RUN AND FAULT SIGNALS TO THE FIRE ALARM PANEL.



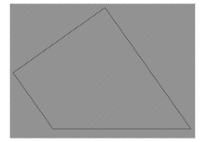
CIVIL DEFENCE

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GENERAL NOTES

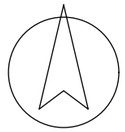
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PROJECT
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PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

FIRE WATER SCHEMATIC

PUMP SET AND RISER DIAGRAM

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

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DATE
18/10/2026

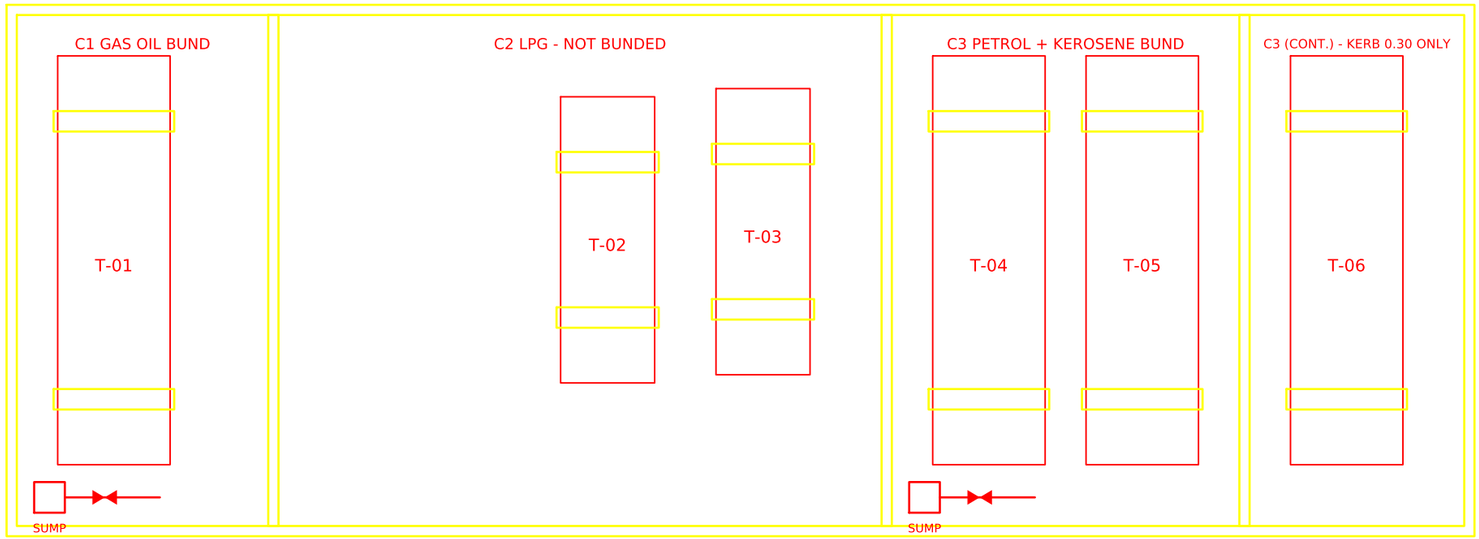
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STATUS
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DRAWING No.
F-006

SHEET
06
OF 8

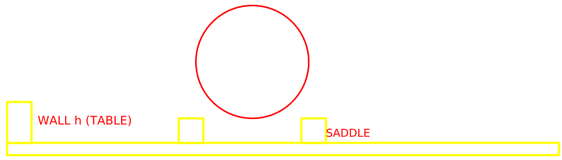
TANK AREA - BUND PLAN 1:150



BUND	TANKS	AREA m2	LARGEST x 1.10	h MIN	WALL h	NET m3
C1	T-01	76.9	59.4	0.82	1.00	70.6
C3	T-04 T-05 T-06	175.0	59.4	0.35	0.60	102.3

1. NET CAPACITY = FLOOR AREA x h - TANK VOLUME BELOW h (TANK BOTTOM 0.60 m ABOVE FLOOR) - SADDLES.
2. REQUIRED: 110 % OF THE LARGEST TANK (NFPA 30 22.11.2 ASKS 100 %); WALL h = h MIN + 0.15 FREEBOARD.
3. C3a AND C3b FORM ONE BUND; THE WALL BETWEEN THEM IS CUT DOWN TO A 0.30 m KERB.
4. LPG COMPARTMENT C2 IS NOT BUNDED: FLOOR FALLS 1 % AWAY FROM UNDER THE TANKS TO AN OPEN GULLY OUTSIDE (NFPA 58 6.4.5 - NO LPG CONTAINER INSIDE A FLAMMABLE LIQUID DIKE).
5. BUND FLOOR 1 % TO THE SUMP; WALLS AND FLOOR LIQUID-TIGHT RC TO ACI 350 (S-005).
6. DRAIN VALVES KEPT CLOSED AND LOCKED; RAINWATER RELEASED ONLY AFTER INSPECTION.

TYPICAL BUND SECTION 1:150



0 1.5 3 4.5 6 7.5m

SCALE 1:150 (A3)

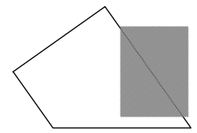
CIVIL DEFENCE

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GENERAL NOTES

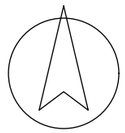
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MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE

BUND COMPARTMENTS

CAPACITY CHECK, DRAIN VALVES, SUMPS

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE 1:150

DATE 18/10/2026

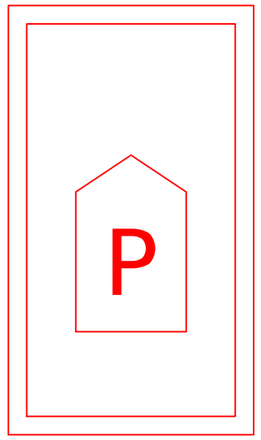
PHASE DESIGN DEVELOPMENT

STATUS FOR REVIEW

DRAWING No. F-007

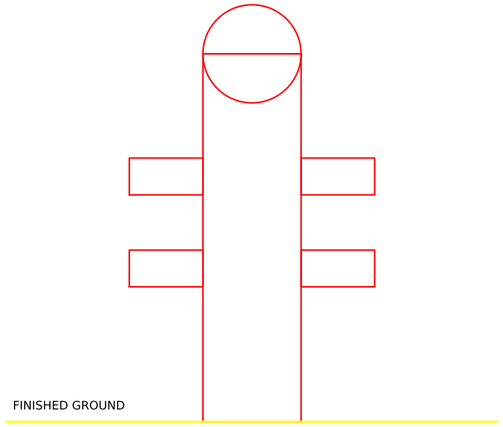
SHEET 07 OF 8

1 EXTINGUISHER CABINET



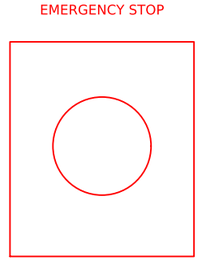
STEEL CABINET, RED, 1.2 mm
GLASS FRONT, TOP <= 1.5 m ABOVE GROUND
PLATE: 'EXTINGUISHER - طفاية حريق'

2 PILLAR HYDRANT



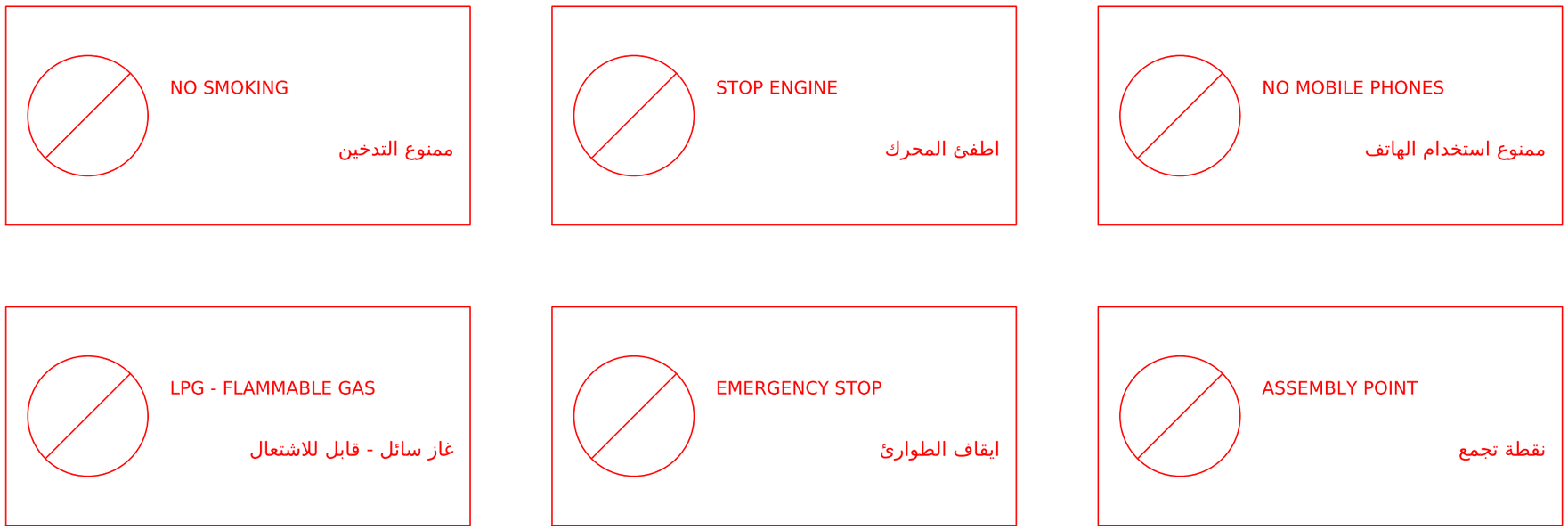
2 x 65 mm OUTLETS, BREAKABLE BASE, DRAIN VALVE; ISOLATING VALVE IN A CHAMBER 1.5 m AWAY; THRUST BLOCK; 0.9 m MIN COVER.

3 ESD PUSH BUTTON



Ex d IIB T4 STATION, RED MUSHROOM, LATCHING ON A POST 1.2 m ABOVE GROUND; ALL BUTTONS IN SERIES IN THE DISPENSER CONTROL CIRCUIT.

4 SAFETY SIGNS (MIN. 600 x 400 mm, REFLECTIVE)



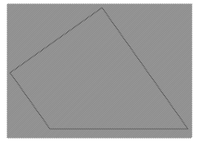
CIVIL DEFENCE

الدفاع المدني

GENERAL NOTES

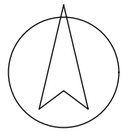
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2. DO NOT SCALE FROM THIS DRAWING. USE FIGURED DIMENSIONS ONLY.
3. VERIFY ALL DIMENSIONS ON SITE BEFORE STARTING ANY WORK.
4. READ WITH THE SITE PLAN AND ALL OTHER DISCIPLINE DRAWINGS AND THE REPORT.
5. COPYRIGHT OF THIS DRAWING IS RESERVED BY THE OFFICE.

KEY PLAN



THIS SHEET

NORTH N



TO BE CONFIRMED

REVISIONS

REV	DATE	DESCRIPTION	BY
00	18/10/2026	FIRST ISSUE - STAGE 0	I.M.

PROJECT
FUEL FILLING STATION
PLOT 1/3453, DISTRICT 43 YARIMJA AL-SHARQIYA

LOCATION
MOSUL, NINEVEH, IRAQ

CLIENT / OWNER
HADI HUSSEIN JUMAA - AHMED OTHMAN EID

PROJECT No.
FS-3453-43

DRAWING TITLE
CIVIL DEFENCE DETAILS
SIGNS, CABINETS, MOUNTING

ROLE	NAME	SIGN.	DATE
DESIGNED	IBRAHIM MAHFOUDH		18/10/26
DRAWN	IBRAHIM MAHFOUDH		18/10/26
CHECKED	-		18/10/26
APPROVED	-		18/10/26

SCALE
NTS

DATE
18/10/2026

PHASE
DESIGN DEVELOPMENT

STATUS
FOR REVIEW

DRAWING No.
F-008

SHEET
08
OF 8